Linear A Detailed Comparative Linguistic Interpretations and Decipherment of a Minoan Cup Inscription

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Abstract

This paper presents three distinct linguistic interpretations of a circular Linear A inscription, derived from comparative methodologies most likely resembling the Linear A writing system: Proto-Hungarian phonetic mapping, Linear B phonetic equivalence, and ancient Egyptian/Coptic semantic analogues. Despite differences in language families and theoretical models, all three interpretations converge thematically on the concepts of light, divine descent, spirit, and celestial origin. This suggests the inscription preserves a ritualistic or poetic message aligned with religious practices in Bronze Age Crete. This study invites archaeologists, epigraphers, and linguists to consider broader frameworks for deciphering Linear A by employing multilingual comparative resonance and modern technological tools.

Introduction

Linear A, the primary script of the Minoan civilization (circa 1800–1450 BCE), remains largely undeciphered.[1] While Linear B, its descendant, was successfully linked to Mycenaean Greek, Linear A has not been connected to the same roots. Scholars debate its origins, proposing connections ranging from Indo-European to Semitic languages, and even hypothetical pre-Indo-European isolates.[2] The Minoans, who lived on Crete, maintained trade networks and cultural contact with other European civilizations. Their labyrinthine architecture, colorful frescoes, palatial wealth, bull worship, and stories of King Minos suggest a society deeply devoted to divine forces.[3] This study analyzes a previously untranslated spiral Linear A

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[1] Jan Best, "The Decipherment of Minoan Linear A," *Talanta* 12–13 (1980–1981): 45–67.
[2] Brent Davis, *Aegean Scripts: Linear A and Cretan Hieroglyphic* (Oxford: Oxford Handbook of Ancient Scripts, 2015).
[3] Arthur Evans, *Scripta Minoa: The Written Documents of Minoan Crete, with Special Reference to the Archives of Knossos*, 28–30 (London: 1909).

inscription using three contrasting linguistic models to explore semantic coherence across interpretations.

The Artifact

The artifact under study is a clay drinking cup with a circular Linear A inscription painted in sepia ink. It was discovered by British archaeologist Arthur Evans in the year 1900 in the foundation of a floor at the Palace of Minos in Knossos, Crete, alongside another painted clay cup and a complete pithos, and is dated to the Middle Minoan IIIa era (ca. 1700 BCE).[4] The cup is a footed goblet—the earliest drinking cup form on Crete, appearing in the mid-third millennium BCE. This goblet shape persisted for centuries, disappearing in the mid-Middle Minoan period (ca. 1900–1800 BCE).[5] Archaeologists suggest it was used domestically or ceremonially.[6] Linear A is typically written from left to right, top to bottom. However, this inscription is arranged in a clockwise spiral of thirteen distinct symbols. Its visual similarity to inscriptions on Minoan gold rings (e.g., the Mavrospilio Ring) and Greek kylixes, as well as its resemblance to later Aramaic incantation bowls, suggests the text may contain incantatory or sacred meaning.[7]

Methodology

The signs were extracted visually and analyzed using three separate decoding systems: Proto-Hungarian phonetic mapping, Linear B phonetic equivalence, and Egyptian/Coptic

[4] Ibid.
[5] Ibid.
[6] Steven Roger Fischer, *History of Writing* (London: Reaktion Books, 2001).
[7] Georgia Flouda, "Materiality of Minoan Writing: Modes of Display and Perception," in *Writing as Material Practice*, 45–67 (London: Ubiquity Press, 2020).

semantic analogues. The comparative values of these 13 symbols are summarized in the following table.

Table: Comparative Values of the 13 Linear A Signs on the Minoan Cup

Order	Linear A Sign*	HT No.	Proto-Hungarian (Rovás)	Linear B (Mycenaean)	Coptic (Egyptian)
1	**✚ cross**	**HT 13**	**□ (T)**	**□ (TA)**	ⲧ **(T)**
2	**□ forked**	**HT 37**	**□ (SZE)**	**□ (SE)**	ⲥ **(S)**
3	**□ comb**	**HT 27**	**□ (FE)**	**□ (FA)**	ⲫ **(Ph)**
4	**□ rectilinear**	**HT 01**	**□ (KE)**	**□ (KE)**	ⲕ **(K)**
5	**□ vertical**	**HT 31**	**□ (TI)**	**□ (TI)**	ⲧⲓ **(Ti)**
6	**□ branch**	**HT 21**	**□ (JE)**	**□□ (SI)**	ⲓⲉ **(Ie)**
7	**□ loop**	**HT 47**	**□ (LLO)**	**□ (LO)**	ⲗ **(L)**

8	**□ M-shape**	**HT 04**	**□ (M)**	**□ (MA)**	**ⲙ (M)**
9	**□ var.**	**HT 37b**	**□ (CSI)**	**□̄(SA)**	**ⲥⲓ (Si)**
10	**□ arch**	**HT 13b**	**□ (E)**	**□ (E)**	**ⲉ (E)**
11	**□ repeat**	**HT 27b**	**□ (SZE)**	**□ (SE)**	**ⲥ (S)**
12	**□ N-form**	**HT 56**	**□ (N)**	**□ (NE)**	**ⲛ (N)**
13	**□ two-pronged**	**HT 17**	**□ (KA)**	**□ (KA)**	**ⲕ (K)**

Interlinear Sequence (All 13 in Three Scripts)

- Proto-Hungarian (rovás): □ □ □ □ □ □ □ □ □ □ □ □ □
- Linear B (Mycenaean): □ □ □ □ □ □□ □ □ □̄□ □ □ □
- Coptic: ⲧ ⲥ ⲫ ⲕ ⲧⲓ ⲓⲉ ⲗ — ⲙ ⲥⲓ ⲉ ⲥ ⲛ ⲕ

Proto-Hungarian Phonetic Mapping

The choice of Proto-Hungarian as a comparator for Linear A is motivated by recent computational linguistic studies suggesting potential links between Uralic and Aegean languages.[8] It is known that trade and cultural exchanges have occurred between the Ancient Carpathian and Aegean regions. Thus, linguists applied feature-based similarity measures across scripts to develop a phonetic grid for Linear A, producing a Minoan-Uralic lexicon and enabling tentative translations of several Linear A inscriptions. Parallel noun case structures identified between Hattic and Hungarian further support a possible West-Ugric linguistic affiliation.[9] While these findings remain unofficial and other hypotheses linking Linear A to languages such as Luwian, Hittite, or Ancient Egyptian exist, the Proto-Hungarian comparison provides a structured framework for exploring phonetic and grammatical correspondences in the Aegean context.[10]

Symbolic Sequence (clockwise):

t – sze – fé – ke – ti – je – llo – m – csi – é – sze – n – ka

Hungarian-style Phonetic Form:

Te szeféketi jellom csé széneka

Translation:

"Light and beauty shine on me; stars and love descend as a sign."

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[8] P. Z. Revesz, *Establishing the West-Ugric Language Family with Minoan, Hattic, and Hungarian by a Decipherment of Linear A,* ResearchGate, 2019.
[9] A. Nepal. *Minoan Cryptanalysis: Computational Approaches to Deciphering Linear A and Assessing Its Connections with Language Families from the Mediterranean and the Black Sea Areas.* Information 15 (2): 73, 2024.
[10] Gábor Varga, *Breakthrough Decipherment of Minoan Linear A*, YouTube Lecture Series, 2024.

This interpretation evokes a spiritual invocation to a divine mother, celestial force, or principle of beauty and light. It aligns with Minoan iconography, where goddesses, celestial motifs, and spirals symbolize life-giving energy.[11]

2. Linear B Phonetic Equivalence

The Mycenaean roots selected for comparison were chosen according to criteria of phonetic compatibility, semantic coherence, cross-linguistic support, and functional relevance to the hypothesized Linear A readings. For each Linear B syllable, preference was given to glosses whose phonetic structure closely corresponded to the proposed Linear A values, whose meanings aligned with the recurring contexts of the inscriptions, particularly administrative, cultic, or economic terms, and whose significance was reinforced by parallels in related Indo-European or Uralic languages. Glosses not selected, while attested in the Mycenaean corpus, often represented rare, contextually isolated, or semantically divergent meanings that were unlikely to reflect the intended usage in Linear A. This methodical selection ensures that the comparative analysis maintains both phonological plausibility and semantic consistency, providing a robust framework for reconstructing the Linear A lexicon.[12]

Transcription (Linear B style):

TA – SE – FA – KE – TI – SI – LO – MA – SA – E – SE – NE – KA

Possible roots from known Mycenaean/Minoan vocabulary:

- TA-SE = *tasija* (ritual/religious term)

[11] Jan Best, "The Decipherment of Minoan Linear A," 50–51.
[12] Cyrus H. Gordon, "Evidence for the Minoan Language," *Journal of Near Eastern Studies* 24, no. 4 (1965): 301–317.

- FA-KE-TI = *phageti* (eats) or *phake* (torch/light)
- SI-LO-MA = *silo* = grain; *ma* = mother
- SA-E-SE = *sa* (holy), *ese* = divine
- NE-KA = *neko* = to die/pass or *nek-ara* (offering)

Translation:

"Torch-lit offering to the grain-mother; the sacred one descends."

This emphasizes light, divine descent, and a goddess or grain-mother, possibly connected to Demeter. Later rituals connecting to Demeter, such as those of the Eleusinian Mysteries, suggest the use of sacred beverages like the psychoactive Kykeon to facilitate spiritual experiences.[13]

3. Egyptian/Coptic Semantic Analogue

Using Egyptian Coptic as a semantic analog offers a significant advantage for translating Linear A, as it preserves both phonetic continuity and a rich morphological and lexical record from earlier Egyptian stages. This approach emphasizes meaning-based correspondences, allowing Linear A signs to be linked to Coptic terms with similar semantic functions, even when precise phonetic values are uncertain. It is particularly effective for administrative, religious, and material vocabulary, where Coptic retains conceptual patterns parallel to those of the Aegean Bronze Age. By grounding translations in semantic analogs rather than solely in phonetic

[13] Ibid.

conjecture, this method enhances the plausibility of proposed readings and provides a structured framework for interpreting otherwise opaque Linear A inscriptions.[14]

Mapped Elements:

- t = feminine article
- sze/fé = shine
- ke/ka = ka (life force)
- ti = to give
- je = that/who comes
- csi = star
- n = to/for

Translation:

"She who shines gives the spirit; from the stars comes light for the soul."

This interpretation emphasizes a divine feminine force, with cosmic light integrating into the consumer's being. The goddess blesses the participant, highlighting hope, salvation, and spiritual understanding.[15]

Limitations

A central limitation of employing cross-language semantic analogs, such as Coptic and Proto-Hungarian, is the risk of confirmation bias, since there is a tendency to favor translations

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[14] Antonio Loprieno, *Ancient Egyptian: A Linguistic Introduction* (Cambridge: Cambridge University Press, 1995); Bentley Layton, *Coptic in 20 Lessons* (Leuven: Peeters Publishers, 2007).
[15] Ibid.

that fit preexisting hypotheses about Linear A meanings. To address this, each proposed reading was systematically compared with all attested Linear B glosses for the corresponding syllables, evaluated against contextual clues within the Linear A inscriptions, and assessed for phonological plausibility and semantic coherence. Alternative readings, including rarer or contextually divergent glosses, were documented and considered, and only those translations that consistently aligned across multiple comparative lines of evidence, including machine-learning conclusions, were retained.[16] This careful cross-validation minimizes interpretive bias while acknowledging the inherently tentative nature of reconstructing Linear A vocabulary.

Discussion

Despite divergent phonetic systems and cultural contexts, all three interpretations share thematic elements: light, a divine or spiritual agent, descent or arrival from above, and a life force or soul. This convergence suggests that the artifact served as a conduit for drinking and offering rituals invoking divine blessings. Even speculative translations reveal enduring archetypal meanings that reflect Minoan nature-oriented spirituality and the desire to connect mortal and immortal realms.[17]

Conclusion

The study demonstrates that comparative, cross-linguistic approaches can offer meaningful insights into Linear A inscriptions. While absolute decipherment remains elusive,

[16] P. Manoj, and Francesco Perono Cacciafoco. "Minoan and the Machines: Computational Approaches to the Decipherment of Linear A." *Annals of the University of Craiova: Series Philology*, *Linguistics* 46 (1-2): 154–164, 2024.
[17] Charlotte Van der Voort, "Inscribed Cups," *Minoan Drinking Cup: Something Clay, Something Old, Something New*, Things That Talk.

Proto-Hungarian, Linear B, and Egyptian/Coptic analyses converge on themes of divinity, celestial power, and ritualistic practice. Future research should combine multilingual comparative models with technological tools to further investigate Minoan script and its cultural significance.

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